GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

INSTRUCTIONS
GEK- 41277A

HARDWARE MAINTENANCE MODULE

PART NUMBER IC3607D201

REV. 1 1/12/77	REV. 2	APPROVALS D. E. Morehart	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	HARDWARE MAINTENANCE MODULE	
ISSUED 2/27/76	PRINTS TO			277A4901 CONT. ON SH. 1	SH NO 0
MADE BY KK COOPER					

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REVISED INFORMATION

Textural information changed on the most recent revision is flagged by a vertical bar along the left hand margin.

I. DESCRIPTION AND FUNCTIONS

1-1. DESCRIPTION OF EQUIPMENT. The Hardware Maintenance Module (HMM) is an optional, plug-in unit used in conjunction with Directo-Matic* Logic Controller(s) for programmable, automated, industrial applications. Although this manual describes the plug-in module, a similar unit, when ordered, can also be supplied as a built-in module and door-mounted in front of the Directo-Matic*Logic Controller (DLC) in the same control cabinet. Functionally, both the door-mounted and plug-in HMM's are identical and the same operating instructions apply. Figure 1-1 shows a front panel layout of the HMM.

1-2. CONNECTION OF HMM INTO SYSTEM. Each Hardware Maintenance Module consists of the following major assemblies:

1-HMM enclosure containing circuit cards and devices
1-Interconnecting cable
1-Hardware Maintenance Interface (HMI) card (IC3606YHMA1)

A. The HMM is attached by cable to the Hardware Maintenance Interface card which plugs into slot F of the DLC card rack. This connection is shown in Figure 1-2.

CAUTION

The HMM contains no internal power supply. Ensure main power switch on the DLC power supply (see Figure 1-3) is turned off before inserting or removing the Hardware Maintenance Interface card to prevent damage to the HM1 card or DLC malfunction.

1-3. HMM CAPABILITIES. The Hardware Maintenance Module operates through the DLC internal bus to provide the capability of communicating with the sequencer, memory, and other modules for hardware diagnostic purposes. It also provides a powerful capability for software diagnostic uses in systems which make use of the extended capabilities of the DLC (Category II Programming). The HMM can be used for the following functions:

A. Diagnostic Aid. When used with diagnostic software, the HMM furnishes status words from the software to the user, alerting him to specific errors found and the status of a current test undergoing execution.

B. Hardware Maintenance Aid. Allows the user to display and manipulate all registers in the DLC sequencer, and all memory locations and registers in other modules. The user can also start, stop, and single-step the execution of instructions.

C. Software Development Aid. Allows the user to troubleshoot or "debug" programs via control switches and displays. This permits him to display and change the contents of all registers and memory locations. The user can also single-step through a program for troubleshooting purposes, or halt the program on any reference to a pre-selected address.

D. Operational Controller. With the Function Selector in PROGRAM position, special software can make use of the HMM switches and indicators for process operator interface.

* Trademark of General Electric Company, U.S.A.

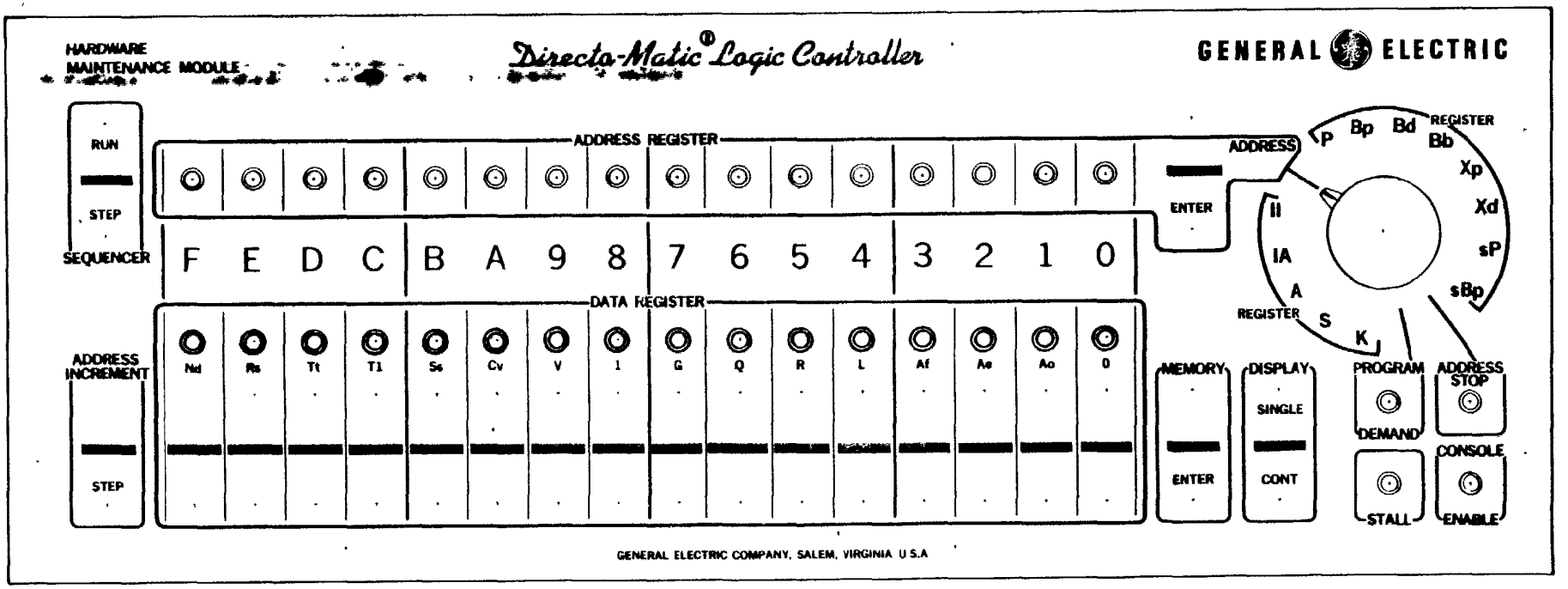


FIGURE 1-1. FRONT PANEL LAYOUT OF HMM

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GENERAL ELECTRIC SALEM, VA. U.S.A.		HARDWARE MAINTENANCE MODULE	
		277A4901	SH NO. 4
		CONT. ON SH. 5	

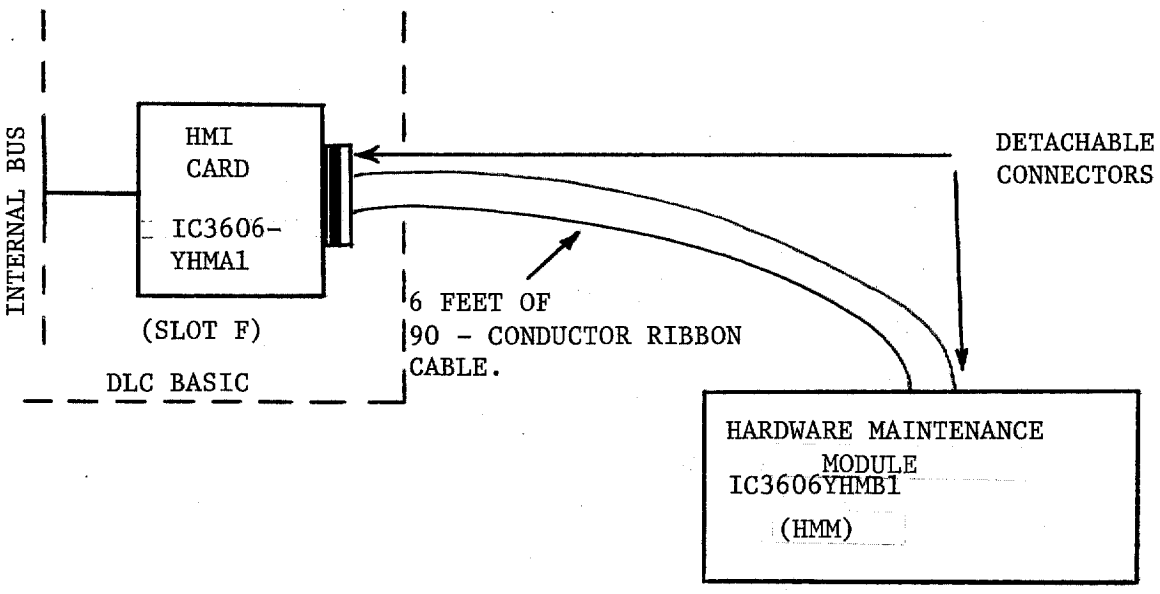


Figure 1-2. HMM Connection to DLC.

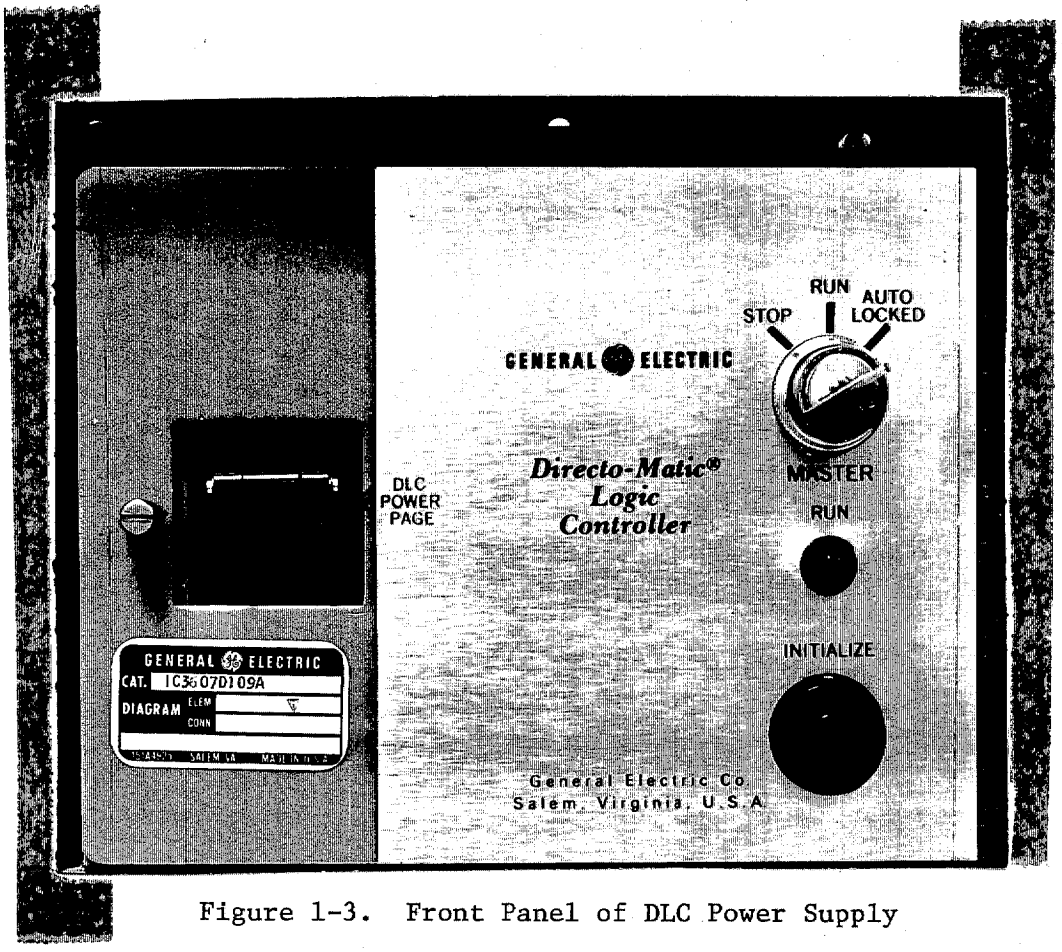


Figure 1-3. Front Panel of DLC Power Supply

REV. 1 1/12/77	REV. 2	APPROVALS D. E. Morehart	GENERAL ELECTRIC	HARDWARE MAINTENANCE MODULE	
ISSUED Feb. 27, 1976	PRINTS TO		SALEM, VA. U.S.A.	277A4901	SH. NO. 5
MADE BY KK Cooper				CONT. ON SH. 6	

1-4. With the Function Selector switch in the PROGRAM position, the DEMAND pushbutton and indicator, the ADDRESS REGISTER indicators and the DATA REGISTER indicators and switches are available for use in a Category II program. The DEMAND pushbutton and indicator are available for synchronizing operator inputs with Category II Code. These registers are available as a virtual memory block beginning at address @EFF0. The functional use of the HMM depends on the design of the Category II routine.

1-5. HMM SWITCHES AND LIGHTS. The capabilities of the HMM will be further discussed by describing the operation of switches and lights mounted on the HMM front panel. Refer to Figure 1-1 for the HMM front panel layout.

A. INITIALIZE AND MASTER (Pushbutton and key switches). The pushbutton and key switches are located on the optional doormounted HMM, but otherwise are located on the DLC power supply (see Figure 1-3). The MASTER key switch is a three-position switch which selects the DLC operating status as follows:

STOP - DLC does not sequence

HMM operational

RUN - DLC sequences

HMM operational

AUTO LOCKED - DLC sequences

HMM operational only in program mode

B. CONSOLE ENABLE (Light). Indicates that the HMM is enabled, or activated by the MASTER key switch (see Figure 1-3). This light is illuminated when the key switch is in the STOP or RUN position. The remaining switches and indicators, except the STALL indicator, on the HMM are functional only when the HMM is enabled.

C. ADDRESS STOP (Light). Indicates that the sequencer has been stopped by an address match when the HMM Function Selector is in the ADDRESS STOP Position.

D. ADDRESS REGISTER (16 Lights). Displays the contents of the address register.

E. DATA REGISTER (16 Lights and two-position binary switches).

1. The 16 lights display the content of the data register.

2. The 16 two-position binary switches function as follows:

Up - null or digital "zero" position.
Down - produce a digital "one".
These switches are used for data entry into the address register, the selected (or addressed) memory location, or for the address stop function.

F. SEQUENCER (3 - Position Switch)

RUN - (Up position) Starts the sequencer operating normally to execute instructions.

CENTER POSITION - Halts the sequencer at the completion of the current instruction.

STEP - (Down - spring return to center position) Causes the sequencer to execute one instruction for each operation of the switch.

NOTE

When stopped, the sequencer has accessed the next instruction and incremented the P-register. The instruction has been based and/or indexed as appropriate. If the instruction is a NOP or DEL, the NOP/DEL flip-flop has been set; if DEL, the IA has been based.

G. ADDRESS ENTER (2-Position Switch).

Up - Null position

ENTER - (Down-spring return to center) Causes the bit pattern set by the DATA REGISTER (binary) switches to be entered into the ADDRESS REGISTER.

H. MEMORY ENTER (2-Position Switch).

Up - Null position

ENTER - (Down - spring return to center) - Causes the bit pattern set by the DATA REGISTER (binary) switches to be entered into the location specified by the Function Selector. If the Function Selector is in the ADDRESS position, the bit pattern will be entered into the memory cell or register whose address appears in the address register.

I. ADDRESS INCREMENT (2-Position Switch).

Up - Null position

STEP - (Down - spring return to center). Causes the contents of the ADDRESS REGISTER to be incremented by one.

J. DISPLAY (3-Position Switch).

SINGLE - (Up-spring return to center). Causes the contents of the location specified by the Function Selector to be displayed in the data register. The display updates only once for each switch operation. Between switch operations, the data is unchanged.

CENTER - (Null position) The data register is unchanged.

CONT - (Down position). This continuous setting causes the contents of the location specified by the Function Selector to be displayed in the data register. The display is continuously updated at line frequency (50 or 60 Hz).

K. DEMAND (Pushbutton and light).

Pushbutton - Sets a demand flip-flop which can be interrogated by software. Light - Indicates status of the demand flip-flop which can be set or reset by the software in the PROGRAM position of the Function Selector.

L. STALL (Light).

Illumination indicates that a stall condition is detected.

M. Function Selector (Sixteen-position rotary switch).

1. PROGRAM - Places the HMM in the program mode i.e., under program control. In this mode, the DEMAND pushbutton, DATA REGISTER switches and the data and address registers are used to communicate between the operator and a DLC program.

NOTE

The "active" operation of the HMM is disabled in the PROGRAM position and a "not busy" indication is available to the software via the status word (@EFF0/F).

2. K - Selects the counting or K-register for display or data entry (@FFFB).

3. S - Selects the status or S-register for display or data entry. (@FFFF).

4. A - Selects the Accumulator or A-register for display or data entry. (@FFFA).

5. IA - Selects the address portion of the instruction or I-register for display or data entry. (@FFF8).

6. II - Selects the instruction portion of the instruction or I-register for display or data entry. Bits 9 through 0 will be "one's". (@FFF9).

7. ADDRESS - Selects the location specified by the contents of the address register for display or data entry.

8. P - Selects the place counter or P-counter for display or data entry. In this position, a memory enter action will set the NOP/DEL flip-flop in the sequencer and cause the next instruction to be taken from the location entered in the P-counter (@0000).

9. BP - Selects the procedure base register (@0001) for display or data entry.

10. B_d - Selects the data base register (@0002) for display or data entry.

11. B_p - Selects the bit base register (@0003) for display or data entry.

12. X_p - Selects the procedure index register (@0004) for display or data entry.

13. X_d - Selects the data index register (@0005) for display or data entry.

14. sP - Selects the "save" location for P(@0006) for display or data entry.

15. sBp - Selects the "save" location for Bp (@0007) for display or data entry.

16. ADDRESS STOP - Places the HMM in the "address stop" mode. While in this mode, the information on the internal bus is compared with the DATA REGISTER switches in each address phase. If a match is detected, a "stop" signal is sent to the sequencer and the ADDRESS STOP light turns on. The condition is reset by turning the Function Selector to any other position. To keep the DLC P-counter at the "stop" location the SEQUENCER switch is set to the center (halt) position before resetting the ADDRESS STOP condition. The data register display is not meaningful in the ADDRESS STOP position of the Function Selector.

NOTE

ADDRESS STOP should only be used in an off-line mode for program development, test or tracing. When a match is detected, all sequencing stops; further, inputs are inhibited and outputs are set to zero (loss of Strobe Enable). The standard ISSP program has a programmed halt (DMR@0) which causes the DLC to hang up once DBC Strobe Enable is lost. The DLC must be re-initialized to recover. The use of Address Stop (off-line) requires that the stall generation circuits be disabled. This is accomplished by jumpering pins 76 of slots J and K on the backpanel to COM. The jumpers must be removed prior to returning the DLC to on-line operation.

17. P-Register Inhibit (2-Position slide switch). This switch is located inside the HMM with the electronics. It is mounted near the upper right-hand side when the HMM is viewed from the rear. When this switch is set in the normal position (toward front of HMM) the P-register advances normally.

When this switch is set in the "save" position (toward) rear of HMM) the P-register advance is inhibited. This switch requires console enabled and the Function Selector in any position other than PROGRAM.

NOTE

The Function Selector switch may have a mechanical limit stop between the PROGRAM and ADDRESS STOP positions.

II. USE AS A DIAGNOSTIC AID

2-1. GENERAL INFORMATION. The value of the HMM as a diagnostic aid is in its ability to view, on-line, the contents of any memory or register address in the system.

2-2. The display of P-register data on the DATA REGISTER lights is the means by which the HMM indicates the region of memory from which the application procedure is being read. This shows whether the program is operating normally or is hung up or halted.

2-3. An on-line process controller stores values in read-write (scratch) memory which reflect the state of control. In a regulator, this data includes reference, feedback, and error data. Data logging applications may include storage of logged variables. The observation of the data in these memory locations by means of the HMM provides diagnostic information pertaining to both the controller and process.

2-4. PROCEDURE FOR DISPLAYING MEMORY CONTENTS. To display HMM memory contents, proceed as follows:

1. Set HMM Function Selector to ADDRESS position.

2. Set DISPLAY switch to the CONT position.

3. Set desired memory or register address on the DATA REGISTER (binary) switches.

4. Load the selected address into the HMM address register by depressing the ADDRESS ENTER switch then returning it to the null position

NOTE

Upon execution of step 4 (above) the address from the DATA REGISTER switches appears on the ADDRESS REGISTER indicators and the content of the addressed location is displayed on the HMM DATA REGISTER indicators. This display is updated at line frequency (50 or 60 Hz). When the display switch is returned to the null position, the display ceases to update. Each time the DISPLAY switch is raised to the SINGLE position and released, the content of the addressed location is accessed and displayed. It remains on the display until the switch is removed from the null position.

NOTE

Do not attempt to monitor nonexistent addresses. Such action will result in the DLC halting on an internal bus fault. The only method of recovery is to re-initialize.

NOTE

Monitoring some controller registers causes the controller to cycle. Do not use the DISPLAY switch in the CONT position for these registers.

2-5. PROCEDURE FOR CHANGING MEMORY CONTENTS. The contents of the addressed location can be changed as follows:

1. Set data word on DATA REGISTER switches.

2. Depress MEMORY switch to ENTER position.

NOTE

Changing some controller registers causes controller action.

NOTE

The contents of memories or registers should not be changed while programs are running except by someone with knowledge of the program and consequences of the change.

2-6. MONITORING I/O AND INTERNAL STATUS VARIABLES. The diagnosis of a problem area in a control system is greatly aided by the capability of the HMM to observe memory locations in real time. Both the Directo-Matic* Logic Controller and the digital I/O units are typically organized on a system level to maintain external address data in the equivalent internal read-write memory addresses. This means that the data from an input termination unit (ITU) at address @2E on the digital party line will also appear in read-write memory at address @002E. With the Function Selector set to ADDRESS, the DISPLAY switch set to CONT, and @002E in the address register, the HMM will display the state of the @2E input group each 16 milliseconds. In this way, the HMM acts as a "window" into the running process.

2-7. Inputs from devices such as position encoders, analog-to-digital converters, and tracking counters are also stored in read-write memory. The address of these inputs may be found in system documentation.

2-8. Outputs from the Directo-Matic Logic Controller are stored in memory prior to being transmitted to the output termination units (OTU's). Similar to the ITU's, the internal addresses of the OTU's are typically correlated with the external addresses. Category II outputs and dynamic intermediate variables can also be monitored using the HMM. Their locations are described in the system documentation.

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III. USE AS A MAINTENANCE AID

3-1. GENERAL INFORMATION. The HMM can be used to stop the sequencer and observe the operation of individual machine instructions. The contents of registers and memory locations can be observed and modified. Each controller on the internal bus can be observed and cycled under manual control. These capabilities make the HMM a powerful maintenance tool.

3-2. SEQUENCER CHECKS. The HMM can be used to enter programs which provide a time base for oscilloscope checks of sequencer instructions. A convenient test point for synchronizing the oscilloscope is the NOP/DEL flip-flop pin J60. A frequently used pattern is as follows:

The following type of check must be performed with the system off-line since the test program requires the machine's entire capability.

```
          @10    0000     NOP

          @11     ()      INSTRUCTION UNDER TEST*

          @12    C010     BRU@10

   Initialize    P=10     BP=0
```

3-3. USE OF SAVE P SWITCH* A "save P" switch is mounted inside the HMM. In the Save position, the switch inhibits the sequencer from updating the P-counter, forcing the same instruction to run over and over again. This switch is useful for troubleshooting individual sequencer instructions.

* Not to be used for a counter or timer-type instruction.

3-4. INPUT/OUTPUT CHECKS. A comparison of input and output states, observed on the ITU/OTU indicators, with data in the read-write memory is a useful maintenance aid. This technique is used in the I/O trouble-shooting procedure found in system instruction book GEK - 40421. This procedure offers a guide for the following symptoms:

1. Signal from an ITU point not reaching memory.
2. Input signals stuck "on" from an entire digital remote terminal.
3. A single ITU/OTU stick does not respond.
4. A group of ITU and/or OTU's does not respond in a single digital remote terminal.
5. Groups of ITU and/or OTU's do not respond throughout an entire party line system.
6. No ITU or OTU in the entire system is responding.
7. No ITU or OTU in one entire digital remote terminal is responding.
8. Signal in the DLC memory not reaching an OTU point.

3-5. CONTROLLER CHECKS. Since the HMM can read from, and write into any memory location defined in the machine, it is useful in checking the operation of the controllers that communicate with the sequencer via their virtual memory locations. The specific details of controller operation are described in the System Instruction Book GEK-40421. Each controller is characterized by a basic virtual memory block as follows:

@FSS0	Control/Status
@FSS1	Not Used
@FSS2	Not Used
@FSS3	External Address
@FSS4	Data

Similarly, each controller is defined by its basic address @FSS0:

DBC =	@FFE0
ABS #1 =	@FFD0
ABS #2 =	@FED0
BTC =	@FFC0
EAU =	@EFE0

A controller can be exercised and cycled by using the HMM to read from and write to the controllers virtual memory registers. The use of the SINGLE position of the DISPLAY switch can avoid multiple READ operations.

3-6. USE WITH TEST AND VERIFICATION SOFTWARE. A family of hardware test and verification programs are available for troubleshooting the sequencer and other DLC modules. These programs are documented in the IC3609 series of drawings. These programs, together with the HMM, provide useful tools for the maintenance of the DLC.

IV. USE AS A SOFTWARE DEVELOPMENT AID

4-1. GENERAL. The most common use of the HMM for software development is the checking and modification of Category II routines. This is made possible by the ability of the HMM to read and modify memory. The memory content of a routine under test may be scanned by the HMM to examine its contents. Test modifications may be made by the HMM to force the routine into a troublesome path and to modify the program logic to make it function properly. To aid in locating the trouble area, the running procedure can be stopped at any selected point by use of the ADDRESS STOP switch.

4-2. PROGRAM TRACING. In the address stop mode, the DLC will halt when an instruction or data word is fetched from the address set into the DATA REGISTER switches on the HMM. This is a useful aid when attempting to isolate a software error that is producing a bus stall. The stalled bus causes the entire system to be unresponsive to any stimulus including the HMM. This condition can be caused by the software addressing a non-implemented memory location and can only be cleared by re-initializing the DLC. Upon re-initializing the DLC, the ADDRESS STOP switch is set to halt the machine at various points in the procedure, starting at the beginning of the routine and sequentially bracketing the faulty instruction until the error is found.

4-3. The ADDRESS STOP switch is also useful in determining the procedure's path through various logic branches. If the DLC does not stop in a given path, that path is not currently being run. Once it is stopped, the SEQUENCER switch may be returned to the center position and the various registers observed.

4-4. DATA MONITORING. The ADDRESS STOP function is also very useful for determining which of several programs is changing a shared data area. With the address of a data word in the DATA REGISTER switches, the DLC will stop each time a program accesses that address. By examining the P-counter, the user can find out which program has accessed the data word. The II register will indicate whether the instruction is a LOAD or STORE operation.

4-5. PROGRAM MODIFICATION. Category I procedures can be checked by means of the Installation and Maintenance Module, without knowledge of programming details. However, Category II procedures require the use of the HMM to observe and modify machine instructions. This capability of the HMM must be used cautiously because the user is not protected from any errors he might make. The Category II procedures can be successfully modified only by someone who is familiar with the machine instructions and the particular program to be modified.

4-6. With the Function Selector switch in the PROGRAM position, the ADDRESS REGISTER indicators and the DATA REGISTER indicators and switches are available for use by a Category II program. The DEMAND pushbutton and indicator are available for synchronizing operator inputs with the Category II code. These registers are available as a virtual memory block beginning at address @EFF0. The functional use of the HMM depends on the design of the Category II routine.

V. HMM CARE AND MAINTENANCE

5-1. GENERAL. The HMM is protected by a steel enclosure and requires no maintenance during use or storage. After use, the following procedure is recommended:

1. Store the cable and interface card in the provided slot at the back of the HMM.

2. Leave the DATA REGISTER switches and other spring-loaded switches in the center position.

3. Turn the Function Selector to the P-position.

5-2. When not in use, the HMM should be stored in a location free from falling objects and highly toxic chemical gases. Do not store it for extended periods on a cluttered workbench in a busy shop environment. Do not attempt to clean or repair switches. If a switch does not function properly, it should be replaced.

5-3. ENVIRONMENT

1. Operating temperature (ambient): $0^{o}C$ to $55^{o}C$.

2. Storage temperature: $-55^{o}C$ to $85^{o}C$.

3. Humidity (operating and storage): 95% without condensation.

5-4. USEFUL PART (DRAWING) NUMBERS:

1. Hardware Maintenance Console IC3607D201, Sheets 1.0 thru 1.3, 2.0 and 2.1.

2. Console Wiring Diagram 68B945079

3. Elementary Diagram Card IC3606YHMB1 (Box)

4. Hardware Maintenance Interface Card IC3606YHMA (Page)

5. IC3607D201A Portable Console
   IC3607D201B Door-Mounted Console

VI. APPENDIX

6-1. BINARY AND HEXADECIMAL EQUIVALENTS

Binary	Hexadecimal	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

Examples

Binary	Hexadecimal Equivalent
0000 0011	3
0000 0111	7
0011 0010	32
0011 1111	3F
0010 1100	2C

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT

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